Worksheet & Answer Key

Metals low in the activity series, such as zinc and tin, can be extracted from their ores using a process called electrolysis. This involves passing an electric current through a molten mixture of the ore and a suitable electrode, causing the metal ions to deposit on the cathode. The resulting pure metal is then collected and purified further by chemical reactions. Additionally, these metals can also be obtained through smelting, which involves heating the ore in the presence of air to produce the metal.

Word Bank:

electrolysis, molten, cathode, purified, chemicals, heating, air, process

Fill in the blanks:

1. The metals low in the activity series, such as zinc and tin, can be extracted from their ores using a _____.
2. This involves passing an electric current through a _____ mixture of the ore and a suitable electrode.
3. The resulting pure metal is then collected and further _____ by chemical reactions.
4. Additionally, these metals can also be obtained through a type of _____, which involves heating the ore in the presence of _____.
5. The process involves passing an electric current through a _____ mixture of the ore and a suitable electrode.
6. The resulting pure metal is then collected and further _____ by chemical reactions.
7. Metals low in the activity series, such as zinc and tin, can be extracted from their ores using a _____ called electrolysis.
8. The metals are then obtained through a type of _____, which involves heating the ore in the presence of air.

■ Answer Key:

1. electrolysis
2. molten
3. cathode
4. purified
5. chemicals
6. heating
7. air
8. process